Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Simeonsson, Josef B. Ph.D.		
Project Title:	Advanced SERS-Based Sensor for Monitoring Air Quality in Space Cabin Environments		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	joesimeonsson@earthlink.net	**Fax:**	FY 919-806-8250
PI Organization Type:	INDUSTRY	**Phone:**	919-806-8250
Organization Name:	Advanced Monitoring, Inc		
PI Address 1:	4608 Industry Lane		
PI Address 2:	Suite L		
PI Web Page:			
City:	Durham	**State:**	NC
Zip Code:	27713	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	10/01/2001	**End Date:**	12/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1298		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The development of novel air quality sensors based on surface-enhanced Raman scattering (SERS) is proposed for monitoring of primary target pollutants in the spacecraft environment. These studies will demonstrate the feasibility of the SERS-based sensors and will address their analytical capabilities for monitoring selected volatile organic compounds in air. The merits of the proposed sensors include: (i) an all solid state design, (ii) high sensitivity, (iii) high chemical selectivity, (iv) a low impact design in terms of size, power, and utility requirements, and (v) low heat production and no chemical waste production. The key components of the proposed sensors are hybrid SERS substrates that serve as the active sensor element. These substrates consist of metal surfaces coated with self-assembled monolayers (SAMs) of thiol-linked organic compounds. The choice of the thiol-linked compound is critical to the performance of the sensor as it protects the surface of the metal from oxidation and provides a chemically selective phase that can be designed to preferentially adsorb the targeted analyte. In effect, the monolayer acts like a chromatographic stationary phase that		

	enables partitioning of the analytes to the surface depending on the favorability of the chemical interaction. For long-term applications, the sensor substrates can be passively deployed as dosimeters that are retrieved for measurements after exposure. Alternatively, the substrates can be regularly monitored via fiber optic coupling to a SERS spectrometer. The proposed studies will address the analytical capabilities of the sensors for monitoring primary target pollutants at their Spacecraft Maximum Allowable Concentration levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of SERS-based sensors promises to result in greatly enhanced monitoring capabilities for NASA and lead to the development of a new generation of air quality sensors. The ability to design substrates to selectively enhance the sensor response to specific analytes or chemical classes gives the proposed sensor great versatility and strong feasibility for a wide range of measurement applications, including occupational health and safety, industrial process monitoring and environmental monitoring. These sensors are also being developed for potential monitoring of chemical agents and other hazardous chemicals related to homeland security issues. Due to their small size and simple design, the sensors can be implemented either as personal dosimeters or as fixed sensors.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)